# MULTIMODAL AI TUTORING SYSTEM

25-26J-130

Project Proposal Report

Loku Bahithage Sachini Kaushalya

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

# MULTIMODAL AI TUTORING SYSTEM

25-26J-130

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

# DECLARATION OF THE CANDIDATE & SUPERVISOR

I declare that this is My own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

| Name | Student ID | Signature |
|---|---|---|
| Kaushalya L.B.S. | IT22356772 | |

Statement Of the Supervisor,

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date: 27/08/25

Signature of the co-supervisor: Date:

# ABSTRACT

Intelligent Tutoring Systems (ITS) play a critical role in the personalization of learners however, most current systems are reactive, involving only static, rule based mechanism which does not dynamically change with the cognitive state of the learner or engagement, or performance. This paper proposes a Multimodal AI Tutoring System with all its focus on a multimodal AI powered content creator with an LLM based Content Generator augmented by Retrieval Augmented Generation (RAG) to generate contextually informed, adaptive, and personalized instructional content.

The suggested approach combines pre-trained transformer based language models, agentic prompt engineering, and multimodal learner feedback. The RAG mechanism anchors generated content in external knowledge base, increasing accuracy and relevance and decreasing hallucinations. Computer vision is a mode that records in real time learner attention and affective states, and the system can dynamically adapt to those indicators. A/B testing and other empirical testing will measure the relevance of the content, pedagogical expertise and engagement of the learners and will inform iteration.

It is expected that the final results will be a scalable, modular design capable of creating personalized quizzes, hints, summations, and explanations based on individual goals and preferences in learning. The system is supposed to increase understanding, motivation, and engagement and illustrate how LLMs, RAG, and multimodal feedback can be used in a unified, proactive instructional system. The study will also offer teachings in putting AI based adaptive systems into practice and help design the next-generation intelligent tutoring technologies.

# TABLE OF CONTENT

DECLARATION OF THE CANDIDATE & SUPERVISOR ..... i
ABSTRACT ..... ii
TABLE OF CONTENT ..... iii
LIST OF FIGURES ..... iv
LIST OF ABBREVIATIONS ..... v
1.INTRODUCTION ..... 1
1.1 Background & Literature Survey ..... 2
2. OBJECTIVES ..... 5
2.1 Main Objective ..... 5
2.2 Specific Objectives ..... 5
3. METHODOLOGY ..... 6
4. PROJECT REQUIREMENTS ..... 10
4.1 Functional Requirements ..... 10
4.2 Non-Functional Requirements ..... 11
4.3 User Requirements ..... 12
4.4 System Requirements ..... 13
5. DESCRIPTION OF PERSONAL AND FACILITIES ..... 14
5.1 Personnel ..... 14
5.2 Facilities ..... 14
6. REFERENCE LIST ..... 16

# LIST OF FIGURES

Figure 3.1:Overall System Digram ..... 7
Figure 3.2: Individual System Diagram ..... 8

## LIST OF ABBREVIATIONS

| | |
|---|---|
| ITS | Intelligent Tutoring Systems |
| RAG | Retrieval Augmented Generation |
| AI | Artificial Intelligence |
| LLMs | Large Language Models |
| CV | Computer Vision |
| RL | Reinforcement Learning |
| NLP | Natural Language Processing |
| EduLLMs | Educational LLMs |
| AQG | Automatic Question Generation |
| HPC | High performance computing |
| PI | principal investigator |
| LMS | Learning Management Systems |

# 1.INTRODUCTION

Educational technology has evolved rapidly, with intelligent tutoring systems (ITS) increasingly used to enhance personalized learning. However, most existing ITS remain reactive, responding to errors or requests without strategic foresight. They largely rely on fixed rule based mechanisms that fail to dynamically adapt to learners' academic levels, cognitive states, or engagement patterns.

Recent advancements in Artificial Intelligence (AI) including Large Language Models (LLMs), computer vision (CV), reinforcement learning (RL), and agentic planning offer new opportunities to create proactive and adaptive tutoring systems. By leveraging these technologies, a tutoring system can shift from being reactive to being goal driven, continuously planning and adapting learning strategies in real time.

This research proposes a Multimodal AI Tutoring System that integrates four core components:

1. Agentic AI Core for goal-oriented planning and decision-making.
2. Reinforcement Learning Engine for optimizing instructional strategies.
3. LLM-Driven Content Generator for personalized, adaptive content creation.
4. Computer Vision Attention Monitor for real-time engagement tracking.

Among these, the LLM Driven Content Generator plays a crucial role by delivering context aware quizzes, hints, summaries, and explanations tailored to individual learners. In particular, Retrieval Augmented Generation (RAG) enables LLMs to generate content that is grounded in external knowledge bases, thereby reducing hallucinations and improving learner comprehension [1]. Integrating RAG based LLMs with agentic prompt control and multimodal feedback mechanisms further enhances personalized learning pathways, supports adaptive assessments, and improves overall student engagement.

### 1.1 Background & Literature Survey

The evolution of Intelligent Tutoring Systems (ITS) has been invigorated by advances in Natural Language Processing (NLP) and, notably, Large Language Models (LLMs). Traditional methods for educational content generation such as template based systems have proven limited in adaptability, often failing to produce contextually rich or dynamically personalized instructional materials. The integration of Retrieval Augmented Generation (RAG) now enables LLMs to ground their outputs in real time, domain specific content, significantly reducing hallucinations and improving relevance.

Dong *et al.* demonstrated the practical benefits of RAG in an engineering education context. Their system combined GPT-4 with Vectara for hybrid retrieval and deployed prompt engineering to generate adaptive multiple-choice questions. The model achieved high topic (≈88.5%) and answer relevance (≈88%) and provided immediate feedback and adaptability in question difficulty [1], [2].

While RAG improves content grounding, deploying such systems reliably in educational settings requires robust engineering. VirtualTA exemplified the integration of LLMs into Learning Management Systems (LMS) (e.g., Canvas), providing functionalities like Q&A, flashcards, auto grading, and empathetic dialogue. However, it also surfaced challenges such as PDF parsing limitations, heterogeneous LMS APIs, and evolving LLM versions [3].

Beyond empirical systems, conceptual frameworks for Educational LLMs (EduLLMs) emphasize personalization, ethical considerations, and evaluation rigor. The review highlights the imperative for interpretability, fairness, and robust evaluation frameworks when applying LLMs in educational contexts [4].

Earlier Automatic Question Generation (AQG) methods (e.g., template/keyword omission) were computationally lightweight but lacked context sensitivity and scalability. These baseline approaches underscore the necessity of RAG and LLM integration to produce pedagogically meaningful and diverse content [5].

**Research Gap**

The integration of LLMs within ITS has witnessed significant advancements in recent years, particularly in generating instructional content that supports personalized and adaptive learning. However, a number of critical limitations persist, leaving substantial opportunities for improvement and innovation.

Firstly, current ITS implementations often fail to effectively leverage RAG for producing contextually grounded and real time adaptive content. Without the integration of RAG mechanisms, the outputs generated by LLMs remain predominantly generic, frequently misaligned with external knowledge repositories, and are prone to factual inaccuracies or "hallucinations." This undermines the reliability and pedagogical value of the content presented to learners.

Secondly, agentic prompt engineering a technique that enables proactive planning and dynamic learning pathway generation based on learner profiles, progress, and objectives remains largely underutilized. This omission limits the adaptability of tutoring systems and hinders their ability to deliver instruction that evolves in response to individual learner needs and learning goals.

Thirdly, the iterative refinement of generated content through empirical evaluation is inadequately addressed. Only a limited number of systems incorporate robust validation methods, such as A/B testing, feedback driven optimization, or continuous performance monitoring, to assess and enhance content relevance, clarity, and educational effectiveness over time.

In addition, there is a significant gap in integrating multimodal engagement feedback into the content generation pipeline. While modern computer vision and affective computing techniques can track learner attention, gaze, and emotional cues, these signals are rarely utilized to adapt generated instructional content in real time, thus constraining the potential for responsive and immersive learning experiences.

Finally, existing LLM based tutoring systems lack comprehensive emotional awareness and fine-grained personalization capabilities. Current approaches often fail to dynamically adjust content tone, difficulty level, and pedagogical style to align with learners' emotional states, cognitive load, or personal preferences, reducing the potential for truly individualized learning support.

**Research Problem**

Despite significant advancements in ITS, several critical limitations persist that hinder their effectiveness in delivering personalized and adaptive learning experiences. Firstly, most existing ITS fail to leverage RAG mechanisms for real time, contextually grounded instructional content generation. This shortcoming limits their ability to dynamically incorporate relevant domain knowledge from external or evolving repositories, resulting in content that is often generic and less responsive to individual learner needs.

Secondly, there is inadequate application of agentic prompt engineering techniques, which are essential for proactively tailoring instructional strategies based on learner profiles, progress trajectories, and cognitive states. Current systems often rely on static or minimally adaptive content delivery, thereby reducing opportunities for personalized intervention.

Thirdly, many ITS lack empirical evaluation frameworks, such as controlled A/B testing or iterative feedback loops, to systematically refine and validate the quality, relevance, and pedagogical impact of generated content. Without such mechanisms, the evolution of content generation models remains largely experimental rather than evidence driven.

Moreover, there is a notable absence of integrated approaches that connect NLP based content generation with multimodal learner feedback mechanisms, particularly those utilizing computer vision for detecting engagement, attention, and emotional states in real time. This disconnect results in a one directional content delivery paradigm, where systems provide material without adequately responding to dynamic learner behaviors and feedback.

Collectively, these limitations highlight a pressing need for a unified, adaptive tutoring architecture that integrates RAG for real time knowledge augmentation, employs agentic prompt engineering for personalized learning pathways, incorporates empirical evaluation methods for continuous improvement, and establishes seamless multimodal feedback loops to foster a truly interactive and context sensitive learning environment.

Existing Intelligent Tutoring Systems lack effective integration of RAG, agentic prompt engineering, empirical evaluation, and multimodal feedback, which limits their ability to deliver truly adaptive and personalized learning experiences.

# 2. OBJECTIVES

## 2.1 Main Objective

To design and implement a RAG powered LLM content generator that delivers personalized, contextually grounded, and pedagogically effective instructional content within an adaptive tutoring framework.

## 2.2 Specific Objectives

1. Develop a Retrieval Augmented Generation (RAG) module capable of dynamically fetching and integrating relevant external knowledge to produce accurate and context aware instructional content.
2. Incorporate agentic prompt engineering to enable proactive content planning and adaptive tailoring based on learner profiles, progress, and learning objectives.
3. Implement iterative evaluation mechanisms, such as A/B testing and feedback driven optimization, to continuously refine the quality, clarity, and educational relevance of generated content.
4. Integrate multimodal learner engagement feedback, including attention and affective cues detected via computer vision, to dynamically adjust content delivery for improved learning outcomes.
5. Enhance personalization and emotional awareness in content generation by adapting tone, style, and difficulty according to learners' cognitive states, moods, and preferences.
6. Evaluate system performance and effectiveness in real or simulated learning environments, measuring metrics such as content relevance, learner engagement, comprehension, and satisfaction.

## 3. METHODOLOGY

1. System Overview and Architecture

The proposed Multimodal AI Tutoring System is designed to provide adaptive, proactive, and personalized learning experiences. The system integrates four core components.

1. Agentic AI Core  Responsible for goal oriented planning, decision making, and dynamically adapting learning pathways based on student profiles, progress, and cognitive states.
2. Reinforcement Learning RL Engine Optimizes instructional strategies by evaluating feedback and selecting the most effective pedagogical interventions.
3. LLM-Driven Content Generator (with RAG) Produces context aware quizzes, summaries, hints, and explanations, grounded in external knowledge bases to improve factual accuracy and personalization.
4. Computer Vision Attention Monitor Tracks real time engagement, gaze, and emotional states of learners to adjust content delivery dynamically.

System Diagram

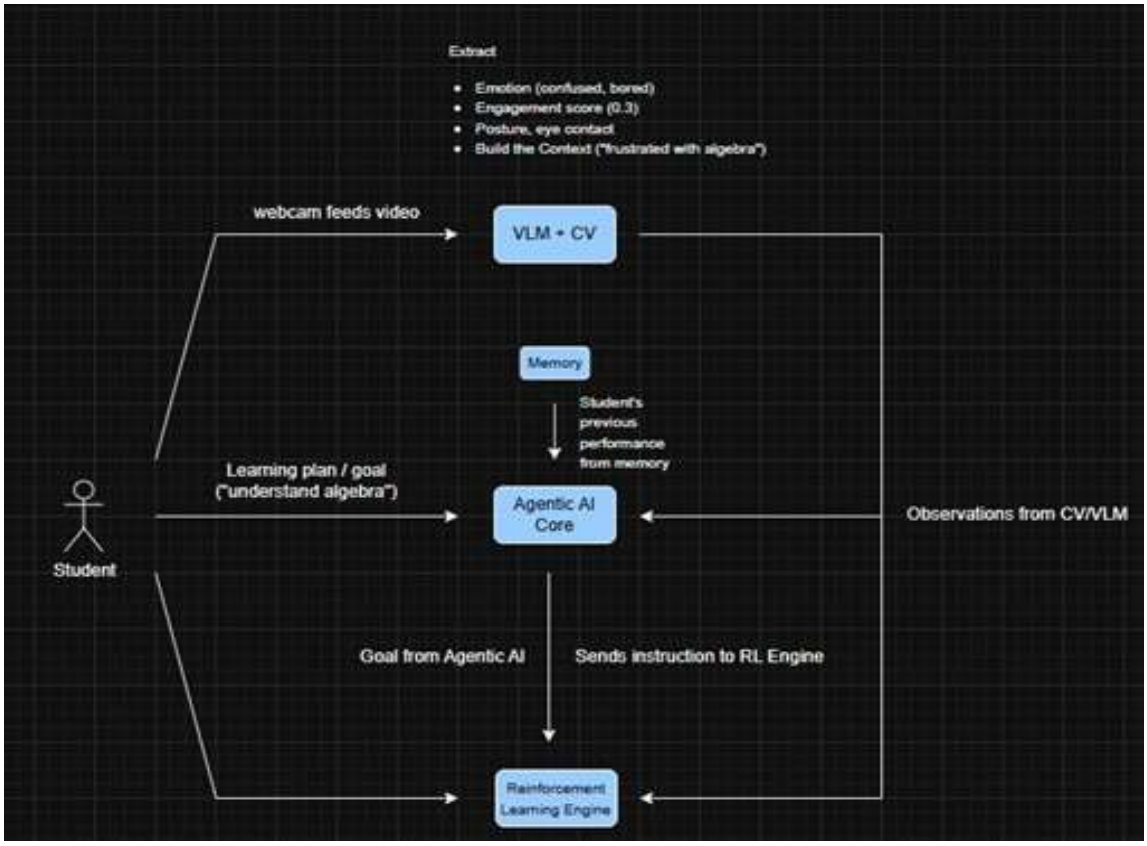


*Figure 3.1:Overall System Digram*

## Individual System Diagram

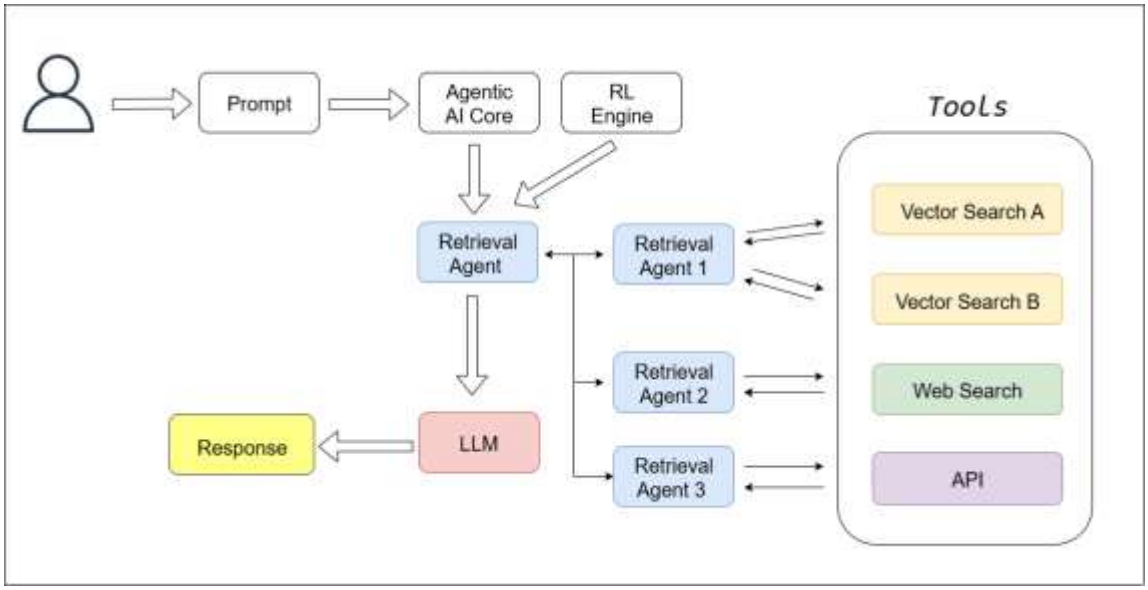


*Figure 3.2: Individual System Diagram*

The proposed research adopts a design, implementation, and evaluation approach to develop a RAG powered LLM content generator, integrating agentic prompt engineering, iterative evaluation, and multimodal learner feedback.

1. System Design and Architecture

- Component Integration: The system will integrate a LLM driven content generator, a RAG retrieval module, an agentic prompt controller, and a learner engagement feedback interface.
- Architecture Overview: A feedback loop will enable the system to generate content, observe learner engagement and responses, and adapt instruction dynamically.
- Data Sources: Educational content will be drawn from textbooks, lecture notes, and domain specific datasets. Semantic vector indexing and hybrid retrieval methods will facilitate real time, contextually grounded content access.

2. RAG Based Content Generation

- Retrieval Module: A hybrid retrieval mechanism, combining semantic and keyword based search, will fetch relevant knowledge to ground LLM generated content.
- LLM Fine Tuning: Pre trained transformer models (e.g., GPT-4) will be fine tuned to generate adaptive instructional content including quizzes, hints, summaries, and explanations.
- Agentic Prompt Engineering: Dynamic, goal oriented prompts will guide the LLM to produce pedagogically aligned content that reflects learner progress, preferences, and objectives.

3. Integration of Multimodal Feedback

- Engagement Detection: Computer vision models will monitor attention, facial expressions, and affective cues to assess real-time learner engagement.
- Adaptive Feedback Loop: Engagement data will influence content generation in real time, adjusting difficulty, tone, and style to enhance motivation and comprehension.

4. Empirical Testing and Iterative Refinement

- A/B Testing: Controlled experiments will compare content variations to evaluate relevance, clarity, and pedagogical effectiveness.
- Performance Metrics: Quantitative and qualitative metrics will measure content accuracy, learner engagement, comprehension, and satisfaction.
- Iterative Optimization: Feedback driven refinement will continuously improve content quality, using insights from learner interaction and performance.

# 4. PROJECT REQUIREMENTS

## 4.1 Functional Requirements

- Content Generation
- Retrieval Augmented Generation (RAG) Integration
    - Retrieve relevant content from a structured knowledge base or vector database.
    - Ensure retrieved content aligns with curriculum and student profile.
- Personalization Engine
    - Adapt content using learner-specific data (e.g., performance history, learning style).
    - Ensure adaptability without forcing emotional or biased personalization.
- Content Validation
    - Automatic validation to ensure generated questions are contextually and technically accurate.
- User Interaction Interface
    - Provide an intuitive interface for students to access quizzes.
    - Allow educators to review and refine generated content.
- Logging and Analytics:
    - Track user engagement and learning outcomes.
    - Provide detailed usage reports for continuous improvement.

## 4.2 Non-Functional Requirements

- Performance
  - The system should generate quiz content within 2 - 3 seconds.
  - Retrieval and RAG integration must operate with minimal latency.
- Scalability
  - Support concurrent access for at least 500 users without performance degradation.
  - Allow easy integration with additional knowledge bases or datasets.
- Reliability & Availability
  - Maintain 99.9% uptime to ensure continuous availability.
  - Implement robust error handling for failed content generation or retrieval.
- Security
  - Ensure secure handling of learner data following data protection regulations.
  - Restrict unauthorized access to educator tools and analytics.
- Usability
  - Provide an intuitive and accessible user interface for students and educators.
  - Ensure compatibility with desktop and mobile platforms.
- Maintainability
  - Use modular architecture for easy updates and integration of new AI models.
  - Provide well-documented APIs for future system enhancements.
- Accuracy & Consistency
  - Ensure the personalization engine delivers contextually appropriate adaptations.
  - Maintain consistency in question quality and difficulty levels across sessions.

### 4.3 User Requirements

1. Ease of Use

The system shall provide an intuitive and user friendly interface for both students and educators, allowing seamless interaction with the content generation, adaptive learning, and feedback modules.

2. Accessibility

The platform shall be accessible through web to support flexible learning environments and ensure inclusivity for all users.

3.Content Relevance

Generated quizzes and instructional materials must align with the curriculum and be tailored to individual learner skill levels, leveraging RAG mechanisms to ensure contextually accurate content.

4.Personalization Needs

The system shall adapt learning materials dynamically according to students' performance history, learning styles, and engagement patterns, incorporating insights from agentic prompt engineering and reinforcement learning modules.

5.Feedback Mechanism

Real time performance feedback shall be provided to learners, including hints, summaries, and explanations generated by the LLM component, to guide and enhance learning outcomes.

6.Educator Control

Educators shall have the ability to review, approve, and modify generated content, ensuring instructional quality and maintaining pedagogical standards while supporting personalized learning pathways.

### 4.4 System Requirements

The proposed system integrates four core components Agentic AI Core, Reinforcement Learning Engine, LLM Driven Content Generator, and Computer Vision Attention Monitor. The system requirements are designed to support the computational, software, and network needs for real time adaptive tutoring.

1. Hardware Requirements

- Processor: Minimum Intel Core i5 or equivalent recommended Intel Core i7 for AI processing and real time engagement tracking.
- Memory (RAM): At least 8 GB RAM 16 GB recommended to handle simultaneous model inference and multimodal data processing.
- Storage: Minimum 500 GB HDD or 256 GB SSD recommended 1 TB SSD for knowledge bases, generated content, and video/visual data storage.
- Graphics: Integrated GPU sufficient dedicated GPU recommended for real time computer vision and reinforcement learning tasks.

2. Software Requirements

- Programming & Frameworks: Python for ML and CV modules, Java or relevant frameworks for backend logic.
- Database: MongoDB or MySQL for storing learner profiles, knowledge base vectors, and interaction logs.
- IDE & Tools: Visual Studio Code or Eclipse, Git for version control.
- AI Libraries: TensorFlow, PyTorch, Hugging Face Transformers, OpenCV.

3. Network Requirements

- Bandwidth: Minimum 10 Mbps internet for knowledge retrieval, cloud based AI inference, and live engagement monitoring.
- Security Protocols: SSL/TLS for secure data transfer; VPN access for remote educator operations.

4. Scalability & Performance

- Support at least 500 concurrent learners and educators without latency in content delivery or feedback.

# 5. DESCRIPTION OF PERSONAL AND FACILITIES

## 5.1 Personnel

The proposed research will be conducted by the principal investigator (PI) with expertise in Data Science, NLP, and AI particularly in transformer based models and educational applications.

1. System Design and Development: Architecting the LLM driven content generator and RAG retrieval framework.
2. Prompt Engineering and Fine Tuning: Designing agentic prompts, fine tuning pre trained LLMs, and implementing adaptive content generation strategies.
3. Multimodal Feedback Integration: Developing and integrating computer vision modules to capture learner engagement and affective signals.
4. Evaluation and Analysis: Conducting empirical experiments, analyzing learner data, and iteratively refining content generation strategies.

Additional support will be provided by research assistants or graduate students, skilled in Python, machine learning, and web based deployment, to assist in:

- Data preprocessing, vectorization, and knowledge indexing.
- Implementation of hybrid retrieval systems and semantic search pipelines.
- Pilot testing and user study coordination.

## 5.2 Facilities

The research will utilize the following facilities:

1. Computational Resources
    - High performance computing (HPC) clusters or GPU enabled workstations for LLM fine tuning and RAG based retrieval.
    - Cloud based AI services (e.g., OpenAI API, Vectara) for scalable deployment and experimentation.

2. Software and Tools
    - Python ecosystem including PyTorch, Hugging Face Transformers, LangChain, NLTK, and SpaCy.
    - Streamlit or similar frameworks for interactive web based interfaces.
    - Computer vision libraries such as OpenCV and MediaPipe for engagement detection.
    - A/B testing and evaluation frameworks for empirical assessment of generated content.
3. Data Resources
    - Domain specific educational corpora including textbooks, lecture notes, and publicly available MCQ datasets (e.g., SciQ).
    - Simulated or real learner interaction data to evaluate adaptive content generation and engagement.
4. Workspaces
    - Laboratory and office spaces for system development, testing, and collaborative meetings.
    - Secure storage for sensitive learner data, ensuring compliance with data privacy standards.

## 6. REFERENCE LIST

[1] C. Dong *et al*., “Enhancing Engineering Education Through LLM-Driven Adaptive Quiz Generation: A RAG-Based Approach,” 2025.

[2] C. Dong, Y. Yuan, K. Chen, S. Cheng, and C. Wen, “How to build an adaptive AI tutor for any course using Knowledge Graph-enhanced Retrieval-Augmented Generation (KG-RAG),” *Proc. 14th Int. Conf. Educational and Information Technology (ICEIT)*, 2025, pp. 152–157.

[3] A. Singh, B. Kumar, and C. Chen, “Artificial Intelligence-Enabled Intelligent Assistant for Personalized and Adaptive Learning in Higher Education,” 2024.

[4] M. Zawacki-Richter *et al*., “Large Language Models in Education: Vision and Opportunities,” *Computers & Education*, vol. 183, 2024.

[5] P. Sharma, R. Mehta, and V. Gupta, “Automatic Question Generation from Textual Data using NLP Techniques,” *Int. J. Educational Technology*, vol. 9, no. 3, 2023.